

Demonstrating Comparing Fabrication Workflows in CAD to Support Design Reasoning

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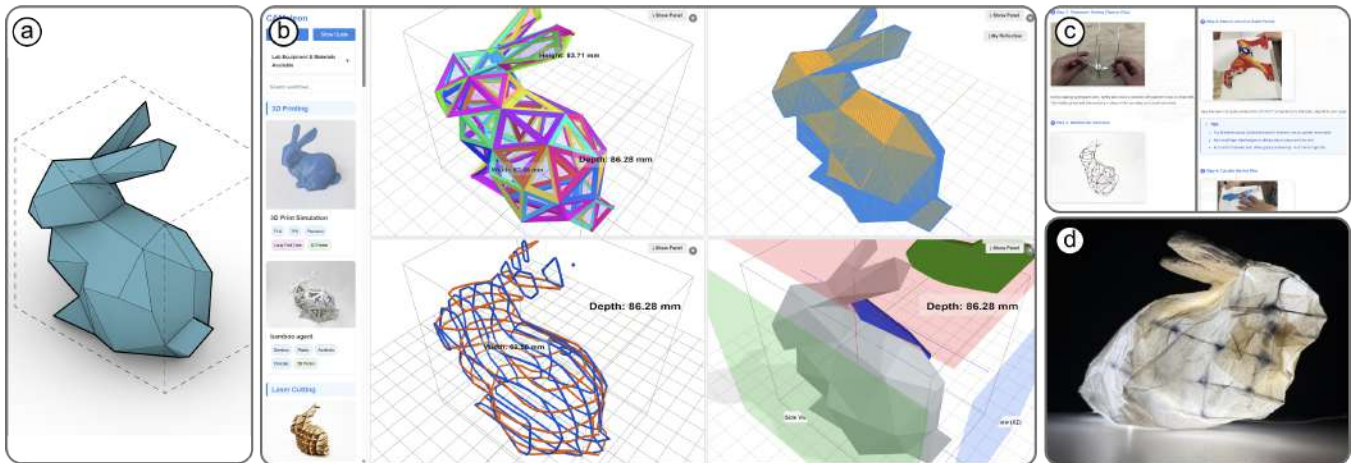


Figure 1: We study the value of comparison in CAD to assist reasoning around fabrication workflows. To this end, we implement an exemplary tool, *CAMEleon*, that enables learners to explore and compare fabrication workflows when initial design assumptions no longer align with their goals. (a) A learner imports a Stanford bunny model into *CAMEleon*. (b) The model displays with side-by-side previews of alternative workflows (e.g., wire forming), highlighting differences in assembly processes, materials, and machine constraints. (c) Learners examine detailed technical information through the workflow information panel for deeper comparison. (d) The final physical outcome (e.g., a bunny lantern) is fabricated using the selected workflow.

Abstract

When novices fabricate, they naturally start by choosing a workflow (e.g., laser cutting, 3D printing, wire bending) and corresponding

software (e.g., Adobe Illustrator, Fusion360, Rhino) from a narrow gamut of options known to them. As they advance their design, another workflow might better suit their design intent, but their models are committed to the original workflow. This prohibits exploration, a learning mechanism fostering informed decision making.



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In this demo, we investigate how to develop CAD interfaces to support exploration and comparison of workflows. We are specifically interested in how such comparison can advance users' reasoning about design decisions. We develop a prototype, we call *CAMEleon*, of such an interface that supports side-by-side comparison of fabrication workflows in CAD. Users load their designed 3D models and preview the fabrication outcomes using various workflows. We hypothesize that presenting alternative outcomes of many workflows supports exploration and scaffolds informed decision-making. When users confirm their workflows, *CAMEleon* lets them export for fabrication and provides instructions on how to manually execute the workflow.

During this hands-on demo at CHI, attendees will use *CAMEleon* to explore different fabrication workflows and reflect on the trade-offs involved in fabrication planning.

CCS Concepts

- **Human-centered computing** → **Interactive systems and tools.**

Keywords

Digital Fabrication, CAD/CAM, Comparison, Design Reasoning, Exploration, Fabrication Education

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1 Introduction

Computer-Aided Design (CAD) and computational fabrication are becoming increasingly accessible to a broader range of users [2]. This trend has enabled diverse applications, such as customized assistive technology [10, 11, 13, 17], DIY robotic devices [19, 20], medical devices [14, 15], and artistic works [7, 12, 16]. As a result, computational fabrication curricula are expanding in educational settings, including middle and high schools [3, 9], maker spaces [4, 6], and graduate engineering and design programs [5, 8, 18].

Through such curricula, learners are well-equipped to get started with a preset of *specific* workflows. For example, a learner may approach a simple box design with 3D printing without considering whether laser cutting or other workflows might better suit their project needs. However, professionals routinely evaluate multiple workflows before committing to one. They assess trade-offs between aesthetics, functionality, and fabricability to reach their design goals [1]. As access to diverse computational fabrication workflows matures, fabrication education faces the challenge of helping learners develop this comparative analysis capability, enabling them to make informed decisions to achieve their goals.

In this demo, we investigate how to support comparison and exploration during CAD design, with the goal of fostering more informed reasoning and decision-making. To study this in practice, we implement an exemplar CAD interface *CAMEleon* that explicitly makes users compare fabrication workflows. The key



Figure 2: Example outputs designed using *CAMEleon* and fabricated using a range of machines, showing how the same model behaves across different fabrication workflows, each with distinct material and structural characteristics.

idea is that users import a CAD model and preview how various workflows would fabricate the same geometry. As shown in Figure 1, users can compare these outcomes side-by-side before deciding which workflow to pursue. The prototype generates previews, provides fabrication metadata (e.g., materials, machines, assembly steps), and highlights workflow-specific considerations (e.g., required post-processing and warnings for possible points of failure). After comparison, users can export models directly from *CAMeleon* to fabricate them in a range of machines, as shown in Figure 2. By making workflows explicitly comparable, we can observe how users explore fabrication workflows by comparing and contrasting them before making decisions, helping us understand how this comparative exploration informs their design reasoning.

2 Walkthrough: Comparing Fabrication Workflows for a Stool

Alex, an undergraduate design student enrolled in a computational fabrication course, wants to create a stool as part of a semester project. However, with only limited exposure to 3D printing and little knowledge of other fabrication workflows, Alex is unsure which workflow is best for the project.

Alex’s instructor recommends using *CAMEleon* to explore fabrication workflows beyond 3D printing. Upon entering *CAMEleon*, Alex imports a stool model generated using *Hyper3D*¹, which generates 3D models from text prompts. (Figure 3a). Alex’s school gives access to the equipment shown in (Figure 3b).

Browsing the sidebar, Alex examines the workflow swatches and keywords for each workflow (Figure 3c). The swatches help Alex understand potential outcomes, while keywords like “durable” and “lightweight” provide guidance on each workflow’s characteristics. Alex initially gravitates toward 3D printing due to familiarity, but decides to explore other options after seeing the diverse possibilities. Alex selects four workflows to compare in detail: epoxy laminating (promising strength characteristics), wire forming (appealing aesthetic), hot wire foam cutting (attractive low-cost foam material), and 3D printing (familiar technology)(Figure 3d).

The side-by-side workflow comparison provides technical information to help Alex decide (Figure 4a). This serves as the core of *CAMeleon*, enabling Alex to evaluate trade-offs across workflows.

¹<https://hyper3d.ai/>

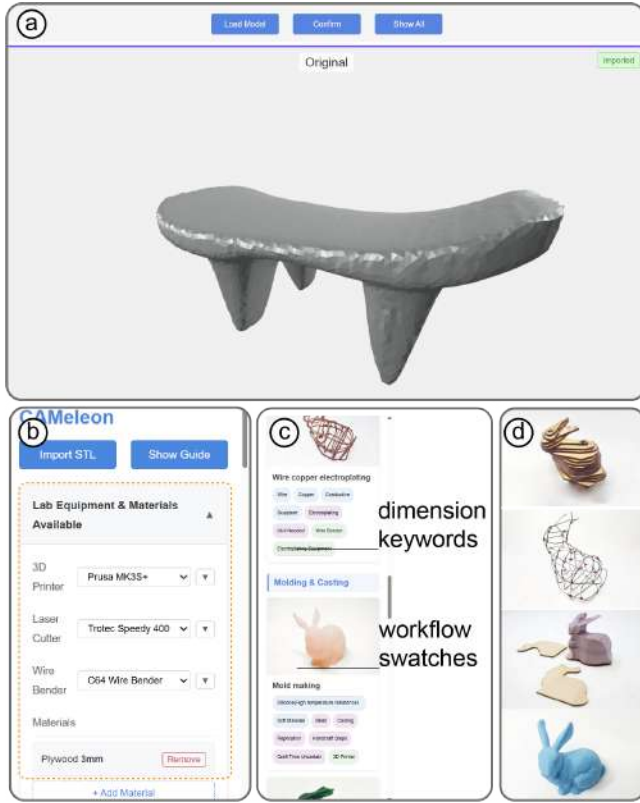


Figure 3: Workflow selection: (a) Alex imports the stool in CAMEleon, (b) workflows are constrained by available tools, (c) sidebar with swatches and keywords for each workflow, (d) four workflows that Alex selects to compare.

For 3D Printing, while Alex is comfortable with the technology, the comparison reveals an estimated print time of 7,479 minutes based on the selected print settings (20% infill, generic PLA) (Figure 4b). CAMEleon warns that the model dimensions may require splitting into multiple parts for the available printer bed size (Figure 4a). Hot Wire Foam Cutting uses inexpensive foam material, with the following guidance note: “Foam strength tip: Standard EPS/XPS foam has limited load-bearing capacity.” Alex considers this note when evaluating the workflow for furniture applications (Figure 4c). Wire Forming displays structural specifications showing that wire construction would create a flexible, springy frame that may not provide the stable seating surface needed for the stool, leading Alex to question this workflow. (Figure 4d). Finally, Epoxy Laminating provides detailed information about material strength, surface finish quality, and the multi-step process involving laser cutting, assembly, and resin application (Figure 4e).

After reviewing the specifications, process complexity, and material properties, Alex chooses Epoxy laminating.

Alex decides to choose the epoxy laminating workflow for the stool. After exporting the required SVG files for the epoxy-laminating workflow, CAMEleon assists Alex step by step with the manual steps to build this stool. Although the guidance note serves as a general framework, CAMEleon provides step-by-step reference instructions

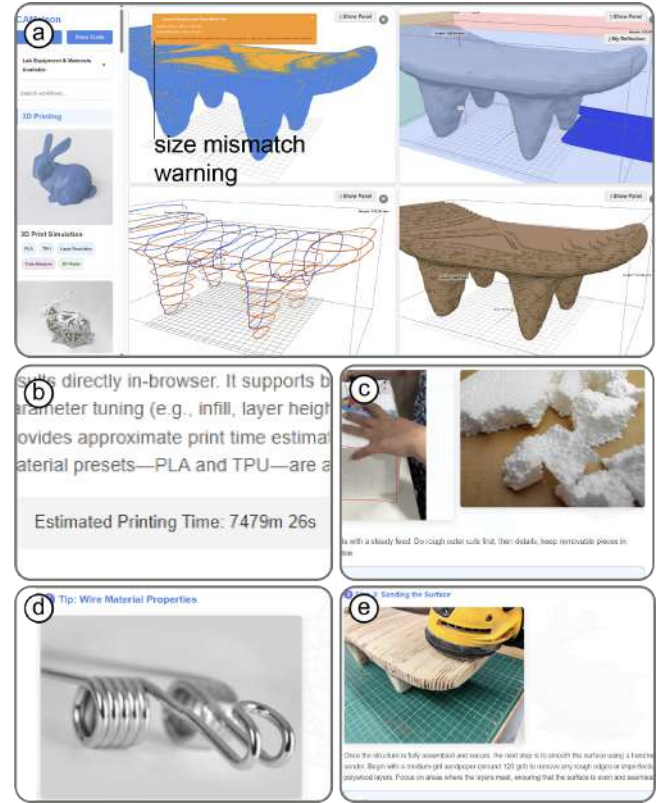


Figure 4: Comparison interface (a) side-by-side view of four candidate workflows, (b) 3D printing workflow showing print time estimates and size warnings, (c) the hot wire foam cutting workflow and assembly process, with material properties indicating limitations, (d) wire forming showing structural specifications and strength characteristics, (e) epoxy laminating presenting strength and finish characteristics.

and videos to outline the typical steps involved in epoxy laminating. Figure 5 illustrates the epoxy laminating workflow guidance.

Through CAMEleon’s comparison, Alex moved beyond 3D printing to adopt a more appropriate fabrication workflow for the project. Besides fabricating the stool using a previously unknown workflow, this exploration experience helps Alex to understand why and when to use such a workflow for future projects.

3 CONCLUSION

This demo investigates how comparison of fabrication workflows can support reasoning during CAD-based planning. Through the design of CAMEleon, we explored comparison as a lens that makes trade-offs, constraints, and alternative fabrication paths more visible to learners. Our lab study provides initial evidence that guided comparison broadens the range of workflows people consider, surfaces conditional reasoning, and helps participants articulate more goal-aligned decisions. These findings point to guided comparison as a promising direction for future CAD tools and fabrication education. Future research should examine how such tools integrate into



Figure 5: Epoxy Laminating workflow execution. (a) Alex laser-cuts the SVG files, (b) assembles and glues the pieces, (c) sands the assembly for surface preparation, (d) applies a layer of epoxy resin to seal and protect the surface, (e) completed stool.

curricula, support long-term learning, and generalize to other fabrication contexts. By foregrounding the reasoning behind fabrication choices, we hope this work contributes toward CAD environments that better support reflective, intentional, and fabrication-aware decision-making.

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References

- [1] Mahdi Alsagour. 2020. Comparing Novice and Expert Designers' Approaches to Design Thinking and Decision Making. 140 pages. <https://www.proquest.com/dissertations-theses/comparing-novice-expert-designers-approaches/docview/2488270158/se-2> Copyright - Database copyright ProQuest LLC; ProQuest does not claim copyright in the individual underlying works; Last updated - 2023-11-20.
- [2] Patrick Baudisch, Stefanie Mueller, and others. 2017. Personal fabrication. *Foundations and Trends® in Human-Computer Interaction* 10, 3–4 (2017), 165–293. Publisher: Now Publishers, Inc..
- [3] Patrick Baudisch, Arthur Silber, Yannis Kommanas, Milan Gruner, Ludwig Wall, Kevin Reuss, Lukas Heilman, Robert Kovacs, Daniel Rechlit, and Thijs Roumen. 2019. Kyub: A 3D Editor for Modeling Sturdy Laser-Cut Objects. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, Glasgow Scotland UK, 1–12. doi:10.1145/3290605.3300796
- [4] Paulo Blikstein. 2013. Digital fabrication and 'making' in education. *FabLab: Of machines, makers, and inventors* (2013), 203–222.
- [5] Leah Buechley, Alyshia Bustos, Eleonora Edreva, Tyler Fenske, Reuben Fresquez, Samuel Hafer, Aislinn Handey, Michelle Louie, John Ng, Alan Shen, Randi Smith, Amber Sustaita, Michael Truong, Kai Vallon, and Kage Micaiah Weiss. 2022. A Computational Fabrication Course: Exploring Philosophical Reflection, Real-World Use, Personal Expression, and Social Connection. In *Proceedings of the 7th Annual ACM Symposium on Computational Fabrication (SCF '22)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3559400.3562006
- [6] Cassia Fernandez, Tatiana Hochgreb-Haeghele, and Paulo Blikstein. 2021. Toward a sustainable model for maker education in public education: Teachers as co-designers in an implementation of educational makerspaces. In *Proceedings of the FabLearn 2020 - 9th Annual Conference on Maker Education (FabLearn '20)*. Association for Computing Machinery, New York, NY, USA, 46–53. doi:10.1145/3386201.3386218
- [7] Marcelo Fraile-Narváez and Mihaela I. Chidean. 2025. Between machines and art: The impact of cnc technology on artistic creation. *Array* (July 2025), 100466. doi:10.1016/j.array.2025.100466
- [8] Neil A. Gershenfeld. 2005. *Fab: the coming revolution on your desktop - from personal computers to personal fabrication* (1. ed ed.). Basic Books, New York, NY.
- [9] Heidi Hartikainen, Leena Ventä-Olkkonen, Marta Cortés Orduña, Ivan Sanchez Milara, Marjukka Käsmä, and Leena Kuure. 2024. Making The Future School: An Analysis of Teens' Collaborative Digital Fabrication Project. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW1 (April 2024), 91:1–91:37. doi:10.1145/3637368
- [10] Erin Higgins, William Berkley Easley, Karen L. Gordes, Amy Hurst, and Foad Hamidi. 2022. Creating 3D Printed Assistive Technology Through Design Shortcuts: Leveraging Digital Fabrication Services to Incorporate 3D Printing into the Physical Therapy Classroom: Leveraging Digital Fabrication Services to Incorporate 3D Printing into the Physical Therapy Classroom. In *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '22)*. Association for Computing Machinery, New York, NY, USA, 1–16. doi:10.1145/3517428.3544816

- [11] Erin Higgins, Zaria Oliver, and Foad Hamidi. 2023. Towards a Social Justice Aligned Makerspace: Co-designing Custom Assistive Technology within a University Ecosystem. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '23)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3597638.3608393
- [12] Stephen Hoskins. 2018. *3D printing for artists, designers and makers* (second edition ed.). Bloomsbury Visual Arts, An imprint of Bloomsbury Publishing Plc, New York.
- [13] Amy Hurst and Jasmine Tobias. 2011. Empowering individuals with do-it-yourself assistive technology. In *The proceedings of the 13th international ACM SIGACCESS conference on Computers and accessibility*. 11–18.
- [14] Swati Kharat, Soumya Iranna Dudhani, Afreen Kouser, Ponniah Subramanian, Debarshi Bhattacharjee, and Vikram Jhamb. 2024. Exploring the Impact of 3D Printing Technology on Patient-Specific Prosthodontic Rehabilitation: A Comparative Study. *Journal of Pharmacy & Bioallied Sciences* 16, Suppl 1 (Feb. 2024), S423. doi:10.4103/jpbs.jpbs_643_23
- [15] Hana Beyene Mamo, Marcin Adamiak, and Anil Kunwar. 2023. 3D printed biomedical devices and their applications: A review on state-of-the-art technologies, existing challenges, and future perspectives. *Journal of the Mechanical Behavior of Biomedical Materials* 143 (July 2023), 105930. doi:10.1016/j.jmbbm.2023.105930
- [16] Bridgette Mongeon. 2016. *3D technology in fine art and craft: exploration of 3D printing, scanning, sculpting and milling*. Focal Press, Burlington, MA Abingdon, Oxon.
- [17] Jeremiah Parry-Hill, Patrick C. Shih, Jennifer Mankoff, and Daniel Ashbrook. 2017. Understanding Volunteer AT Fabricators: Opportunities and Challenges in DIY-AT for Others in e-NABLE. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17)*. Association for Computing Machinery, New York, NY, USA, 6184–6194. doi:10.1145/3025453.3026045 event-place: Denver, Colorado, USA.
- [18] Kanjun Qiu, Leah Buechley, Edward Baafi, and Wendy Dubow. 2013. A curriculum for teaching computer science through computational textiles. In *Proceedings of the 12th International Conference on Interaction Design and Children (IDC '13)*. Association for Computing Machinery, New York, NY, USA, 20–27. doi:10.1145/2485760.2485787
- [19] Angelique Taylor, Tauhid Tanjim, Michael Joseph Sack, Maia Hirsch, Kexin Cheng, Kevin Ching, Jonathan St George, Thijs Roumen, Malte F. Jung, and Hee Rin Lee. 2025. Rapidly Built Medical Crash Cart! Lessons Learned and Impacts on High-Stakes Team Collaboration in the Emergency Room. arXiv:2502.18688 (Feb. 2025). doi:10.48550/arXiv.2502.18688 arXiv:2502.18688.
- [20] Agisilaos G Zisimatos, Minas V Liarakis, Christoforos I Mavrogiannis, and Kostas J Kyriakopoulos. 2014. Open-source, affordable, modular, light-weight, underactuated robot hands. In *2014 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 3207–3212.